
Statistical Machine Learning – Quiz 3

CSE342/CSE542/ECE356/ECE556

Total marks # 10

Instructions

- Use clear notation and state any assumptions you make.
- Closed-form solutions may not be feasible in some cases.

Course Outcomes

CO1	Students will be able to understand the various key paradigms for machine learning and pattern classification
CO2	Students will be able to apply suitable feature extraction and classification technique to solve a given pattern classification problem
CO3	Students will be able to design a complete machine learning/pattern classification algorithm and evaluate the performance

Questions

Q1. Consider a highly imbalanced binary classification problem in which very few examples belong to the positive class. Suppose that classifying a positive example as negative is much costlier than classifying a negative example as positive. Among accuracy, precision, and recall, which metric should be prioritized? Justify your answer. You must explicitly mention the formula for the metric to be prioritized. [1] [CO1]

Q2. Consider F1 score of classifier f . Suppose a new classifier f_{new} has double the precision and half the recall of the classifier f . Will the F1-score of both classifiers be same? Explain your answer using expression for F1. [1] [CO1]

Q3. In ordinary least-squares regression, one may write the residual as $e = y - f(x; w)$ where $f(x; w) = x^T w$ and assume $e \sim \mathcal{N}(0, \sigma^2)$, which leads to least squares. Now suppose instead that the residual is modeled as $e \sim \text{Bernoulli}(p)$. Derive the likelihood or log-likelihood as a function of w . Find a solution for w . Is this a good model for standard regression? [2] [CO3]

Q4. Ridge regression can be obtained as a MAP estimate using a Gaussian likelihood for the error and a Gaussian prior on the weights. Assume a multivariate Bernoulli prior on weights. Derive the MAP objective function for w . You do not need to solve the optimization problem. [2] [CO3]

Q5. Suppose the true data-generating process is

$$y = f(x) + \epsilon,$$

where the noise is $\epsilon \sim \text{Uniform}(-a, a)$. Derive the bias-variance decomposition of the expected squared prediction error

$$\mathbb{E}[(y - \hat{f}(x))^2].$$

Clearly identify the bias², variance, and irreducible error terms, and compute the irreducible error explicitly for the uniform-noise case. [2] [CO1]

Q6. A dataset is split into two parts: 80% for training and 20% for validation. A student trains polynomial regression models of degrees 1 to 10 on the 80% training part. Following is observed:

Degree 2: training MSE = 8.1, validation MSE = 10.4

Degree 5: training MSE = 4.2, validation MSE = 7.8

Degree 9: training MSE = 1.1, validation MSE = 13.6

Which degree should be selected, and why? Which phenomenon - bias or variance, occurs in other degrees? Explain. [1] [CO2]

Q7. a. A very complex classifier always gives the best test accuracy. True or False? Explain. [.5] [CO1]

b. Regularization usually reduces variance. True or False? Explain. [.5] [CO1]